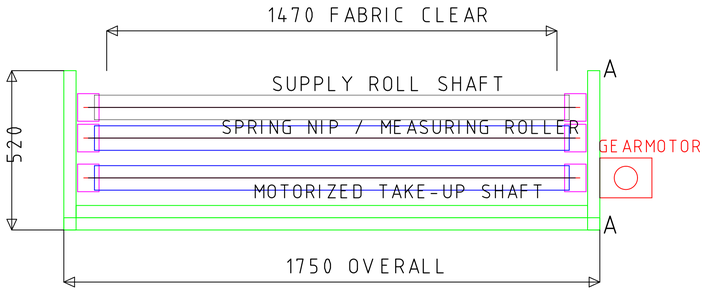
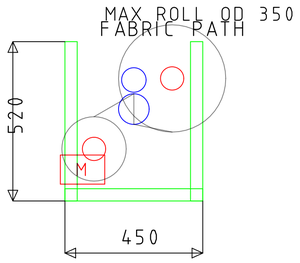
FABRIC ROLL LENGTH METER - GENERAL ARRANGEMENT

REV B MOTORIZED CONCEPT | PROVISIONAL DIMENSIONS | UNITS: mm | SCALE: 1:10 ON A1

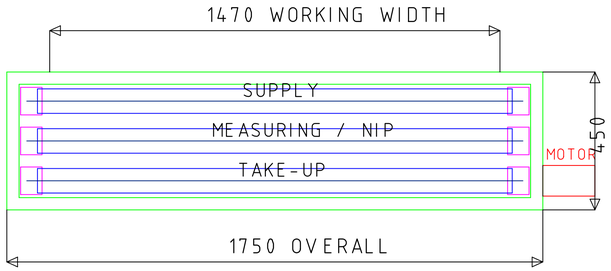
FRONT ELEVATION



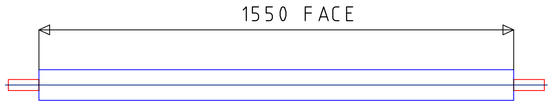
SIDE SECTION A-A



TOP VIEW

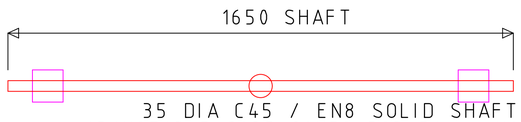


MEASURING ROLLER DETAIL



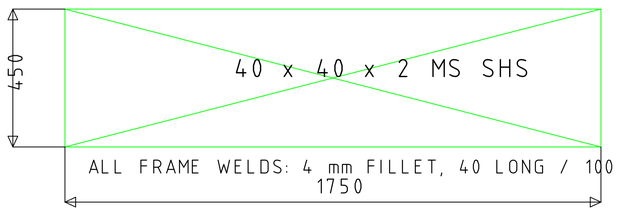
100 OD RUBBER-LAGGED ALUMINIUM ROLLER
35 DIA SHAFT ENDS | ENCODER DIRECT-COUPLED
ROLLER RUNOUT <= 0.20 TIR

SUPPLY / TAKE-UP SHAFT DETAIL



35 DIA C45 / EN8 SOLID SHAFT
ADJUSTABLE CORE CONES FOR 76 mm NOMINAL CORE

FRAME / WELDMENT



ALL FRAME WELDS: 4 mm FILLET, 40 LONG / 100 PITCH UNLESS NOTED

BILL OF MATERIALS

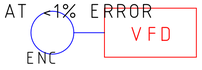
ITEM	DESCRIPTION	QTY	MATERIAL	REMARK
1	BASE FRAME WELDMENT	1	MS SHS	40x40x2
2	SUPPLY SHAFT	1	EN8	35 DIA x 1650
3	TAKE-UP SHAFT	1	EN8	35 DIA x 1650
4	MEASURING ROLLER	1	AL+RUBBER	100 OD x 1550
5	NIP PRESSURE ROLLER	1	AL+RUBBER	80 OD x 1550
6	PILLOW BLOCK UCP207	6	BOUGHT	35 mm BORE
7	ROTARY ENCODER	1	BOUGHT	>=1024 PPR
8	SPRING ADJUSTER	2	MS/SPRING	25-80 N EACH
9	GEARMOTOR + VFD + CLUTCH	1 SET	BOUGHT	0.55 kW PROV.
10	GUARDS / E-STOP / FOOT CTRL	1 SET	MIXED	SAFETY SET

DESIGN / FABRICATION NOTES

1. BASIS: 1470 WIDTH, 40 kg ROLL, 5000 m / 8 h; OD 350 AND CORE 76 PROVISIONAL.
2. SUPPORT SHAFTS IN UCP207 BEARINGS; ALIGN BEARING CENTRES WITHIN 0.5 mm.
3. MEASURING ROLLER CONTACTS FABRIC THROUGH SPRING NIP; PREVENT FABRIC SLIP.
4. VARIABLE LINE SPEED 0-20 m/min; NORMAL PRODUCTION RANGE 12-15 m/min.
5. TARGET <1% REQUIRES ROLLER OD CALIBRATION AND A MEASURED SLIP TEST.
6. FIT E-STOP, HOLD-TO-RUN FOOT CONTROL, TORQUE LIMITER AND PINCH GUARDS.
7. DEBURR; PRIME AND POWDER COAT FRAME AFTER TRIAL ASSEMBLY.
8. GENERAL TOLERANCE +/-1 mm; MACHINED SHAFT FEATURES +/-0.05 mm.
9. VERIFY ROLL OD, CORE, WEB TENSION AND FLOOR SPACE BEFORE RELEASE.
10. PRELIMINARY REV B: FABRICATOR / ENGINEER MUST VALIDATE LOADS BEFORE BUILD.

COUNTER / CALIBRATION

ENCODER: INCREMENTAL A/B, >=1024 PPR, 5-24 V
DISPLAY: 6-DIGIT TOTALISER, RESET + DIRECTION
DRIVE: 0.55 kW GEARMOTOR + VFD, 0-20 m/min PROVISIONAL
CONTROL: E-STOP + HOLD-TO-RUN FOOT PEDAL + SPEED KNOB
CALIBRATE K AFTER 10 m TEST; VERIFY 3 x 20 m AT <1% ERROR



FABRIC ROLL LENGTH METER GENERAL ARRANGEMENT & FABRICATION	REV B UNITS mm	A1 2026-10-16
DRAWING: FRM-1470-GA-001 STATUS: PRELIMINARY / NOT FOR PRODUCTION	SCALE 1:10	OPEN CAD STUDIO